Channel Capacity: Shannon's Second Theorem

Lecture notes on Chapter 7 of Cover and Thomas

Lecture notes for an information theory course
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Notation and standing conventions

Logarithms are base 2 and information is measured in bits; ln denotes the natural logarithm when it appears. Prerequisites from Chapters 2 to 5 of Cover and Thomas [1] are used freely and recalled where they are needed: entropy $H(X)$, conditional entropy, relative entropy $D(p\,\|\,q) \geq 0$, mutual information $I(X;Y) = H(Y) - H(Y|X) = H(X) - H(X|Y)$, its chain rule, its concavity in the input distribution for a fixed channel, the data-processing inequality, Fano's inequality, the asymptotic equipartition property, and the source coding theorem.

Symbol	Meaning (where defined)
$\mathcal{X},\ \mathcal{Y}$	input and output alphabets of a channel, both finite (Session 1)
$p(y\|x)$	transition probabilities of a channel, the entry in row x and column y of the transition matrix (Session 1)
$(\mathcal{X}, p(y\|x), \mathcal{Y})$	a discrete channel; the discrete memoryless channel is written DMC (Session 1)
$C = \max_{p(x)} I(X;Y)$	(information) capacity of a channel, in bits per channel use (Session 1)
$h(p) = -p\log p - (1-p)\log(1-p)$	binary entropy function (Session 1)
$x^n = (x_1, \ldots, x_n)$	a sequence; X^n the corresponding random sequence; $p(y^n\|x^n) = \prod_i p(y_i\|x_i)$ for a DMC without feedback (Session 2)
$W,\ \hat{W}$	message index in $\{1, \ldots, M\}$ and the decoder's estimate (Session 2)
(M, n) code, $x^n(w),\ g$	code with M codewords of length n, the codeword of message w, the decoding function $g : \mathcal{Y}^n \to \{1, \ldots, M\}$ (Session 2)
$\lambda_w,\ \lambda^{(n)},\ P_e^{(n)}$	conditional probability of error given w, maximal error probability $\max_w \lambda_w$, average error probability $\frac{1}{M}\sum_w \lambda_w$ (Session 2)
$R = \frac{\log M}{n}$	rate of an (M, n) code in bits per channel use; $(2^{nR}, n)$ means $(\lceil 2^{nR} \rceil, n)$ (Session 2)
$A_\varepsilon^{(n)}$	set of jointly ε-typical pairs (x^n, y^n) with respect to $p(x, y)$ (Session 2)
$\mathcal{C}$	a codebook, the $M \times n$ array of codewords (Session 3)
C_{FB}	capacity with noiseless feedback (Session 4)
$\mathsf{H},\ d_{\min}$	parity-check matrix and minimum distance of a linear code (Session 5)
(n, k, d)	block length, number of information bits, minimum distance of a linear code (Session 5)
$H(\mathcal{V})$	entropy rate of a stationary source $V_1, V_2, \ldots$ (Session 5)
$W,\ P,\ N_0,\ E_b/N_0$	bandwidth in hertz, signal power, one-sided noise spectral density, energy per bit over noise density (Session 6)

Convention 0.1 *(Standing conventions)*.

(i) A channel is used n times in succession without feedback unless feedback is stated; for a DMC this means $p(y^n|x^n) = \prod_{i=1}^n p(y_i|x_i)$.

(ii) Following Cover and Thomas, a $(2^{nR}, n)$ code means a $(\lceil 2^{nR} \rceil, n)$ code; the ceiling changes the rate by less than $1/n$ and is dropped from the notation.

(iii) "Rate R is achievable" always refers to the *maximal* probability of error $\lambda^{(n)} \to 0$ along a sequence of $(2^{nR}, n)$ codes; Session 3 shows that the average criterion gives the same capacity.

(iv) Capacity without qualification means the information capacity $C = \max_{p(x)} I(X;Y)$, which Sessions 3 and 4 prove equal to the operational capacity; the maximum exists because I is continuous on the compact set of input distributions.

1 Channels and capacity

The first part of the course removed redundancy: a source of entropy rate H can be described with H bits per symbol and no fewer.

This part adds redundancy on purpose. A channel corrupts what is sent, and the question is how many bits per use of the channel can nevertheless be delivered with a probability of error as small as we please.

The answer, Shannon's second theorem, is a single number computed from the channel alone, the *capacity*.

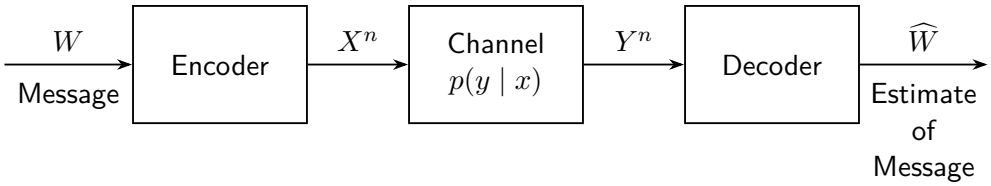
1.1 Discrete channels and the information capacity

Definition 1.1 *(Discrete channel, memoryless channel).* A *discrete channel* $(\mathcal{X}, p(y|x), \mathcal{Y})$ consists of a finite *input alphabet* $\mathcal{X}$, a finite *output alphabet* $\mathcal{Y}$, and for each $x \in \mathcal{X}$ a probability mass function $p(\cdot|x)$ on $\mathcal{Y}$: $p(y|x)$ is the probability of receiving y when x is sent.

Arranged as a $|\mathcal{X}| \times |\mathcal{Y}|$ matrix with rows indexed by inputs, the numbers $p(y|x)$ form the *transition matrix*, whose rows sum to 1.

The channel is *memoryless* if, when it is used repeatedly, the output at each use depends only on the input at that use, and is conditionally independent of earlier inputs and outputs.

We abbreviate *discrete memoryless channel* to DMC.



Communication System. (Figure from Cover and Thomas)

A channel is a conditional distribution and nothing else: it does not know what the sender means, only what symbol is presented to it. If the sender puts a random variable $X \sim p(x)$ into the channel, the output is a random variable Y with joint law $p(x)p(y|x)$, and the mutual information $I(X;Y)$ measures how much the output tells about the input.

The sender controls $p(x)$; the channel controls $p(y|x)$. Capacity is what the sender can get out of the received message.

Definition 1.2 *(Information capacity).* The *information capacity* of a discrete memoryless channel is

$$C = \max_{p(x)} I(X;Y), \tag{1}$$

the maximum over all input distributions $p(x)$ on $\mathcal{X}$ of the mutual information between the input and the output, in bits per channel use.

The word "information" marks that (1) is, for now, only a formula. Sessions 2 to 4 define the *operational* capacity as the largest rate at which messages can be sent with vanishing probability of error, and prove that the two coincide. Once that is done the adjective is dropped. The maximum in (1) exists: $I(X;Y)$ is a continuous function of the vector $p(x)$ on the compact set of probability distributions on $\mathcal{X}$ (Proposition 1.13 below).

1.2 Five channels

The computations use the identity $I(X;Y) = H(Y) - H(Y|X)$:

a channel is good when the output has *large entropy* and
the output given the input has *small entropy*.

Example 1.3 *(Noiseless binary channel).* $\mathcal{X} = \mathcal{Y} = \{0,1\}$ and $p(y|x) = \mathbf{1}\{y = x\}$. Every bit arrives intact, so one bit per use can be sent without error and no more, since only two symbols exist. The formula agrees: $H(Y|X) = 0$, so $I(X;Y) = H(Y) \le 1$, with equality for the uniform input. $C = 1$ bit.

Example 1.4 *(Noisy channel with nonoverlapping outputs).* $\mathcal{X} = \{0, 1\}$, $\mathcal{Y} = \{1, 2, 3, 4\}$, input 0 goes to 1 or 2 with probability $\frac{1}{2}$ each, input 1 goes to 3 with probability $\frac{1}{3}$ and to 4 with probability $\frac{2}{3}$. The output is random but the input is a function of it: outputs $1, 2$ mean 0 and outputs $3, 4$ mean 1. So again every bit is recovered and $C = 1$ bit. In the formula, $H(X|Y) = 0$ gives $I(X;Y) = H(X) \leq 1$, attained by the uniform input.

The point of this example is that noise is harmless when it never makes two inputs look alike: "noisy" and "unreliable" are different things.

Example 1.5 *(Noisy typewriter).* $\mathcal{X} = \mathcal{Y} = \{\mathrm{A}, \ldots, \mathrm{Z}\}$, and each letter is received either unchanged or as the next letter in the alphabet, each with probability $\frac{1}{2}$, with Z wrapping around to A. Using only every other letter, A, C, E, ..., Y, no two inputs can produce the same output, and 13 symbols are sent without error per use: at least $\log 13 = 3.7004$ bits. Conversely, $H(Y|X = x) = 1$ for every x, so $I(X;Y) = H(Y) - 1 \leq \log 26 - 1 = \log 13$, with equality for the uniform input, which makes Y uniform. $C = \log 13$ bits.

This channel is the model for the whole theory. As we will see, every channel, used many times, behaves like a noisy typewriter with 2^{nC} noninterfering inputs.

Example 1.6 *(Binary symmetric channel, BSC).* $\mathcal{X} = \mathcal{Y} = \{0, 1\}$ and each bit is flipped with probability p, independently: $p(1|0) = p(0|1) = p$, $p(0|0) = p(1|1) = 1 - p$.
This is the simplest channel with genuine errors: a received bit is never known to be right. For any input distribution, $H(Y|X = x) = h(p)$ for both x, so

$$I(X;Y) = H(Y) - h(p) \leq 1 - h(p), \tag{2}$$

with equality when Y is uniform, which happens exactly when X is uniform, since then $\mathbb{P}(Y = 1) = \frac{1}{2}(1 - p) + \frac{1}{2}p = \frac{1}{2}$. Hence

$$C_{\mathrm{BSC}}(p) = 1 - h(p) \quad \text{bits.} \tag{3}$$

Numerical examples: $C = 0.9192$ for $p = 0.01$, 0.7136 for $p = 0.05$, 0.5310 for $p = 0.1$, 0.5001 for $p = 0.11$ and 0 for $p = \frac{1}{2}$ (Figure 2).
A channel that flips one bit in ten still delivers more than half a bit per use, with arbitrarily small error, once the right code is used; that codes exist is the content of the result we will show.
Note that: $C(p) = C(1 - p)$: a channel that flips every bit with probability 0.9 is as good as one that flips with probability 0.1, since the receiver can complement what it reads.
Only when $p = \frac{1}{2}$, and the output is independent of the input, the channel is useless.

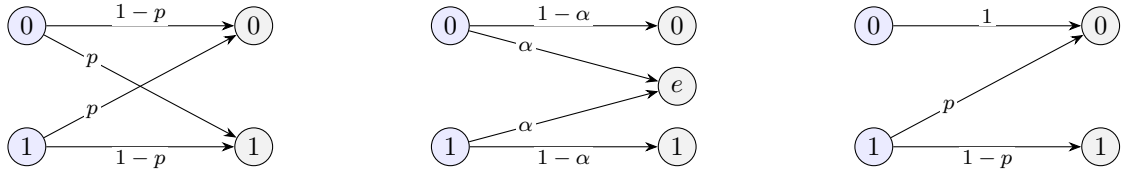
Example 1.7 *(Binary erasure channel, BEC).* $\mathcal{X} = \{0, 1\}$, $\mathcal{Y} = \{0, e, 1\}$, and each bit is either delivered intact, with probability $1 - \alpha$, or replaced by the *erasure* symbol e, with probability α.
Unlike the BSC, the receiver knows which bits were lost.
With $\pi = \mathbb{P}(X = 1)$, $H(Y|X) = h(\alpha)$, and to compute $H(Y)$ let $E = \mathbf{1}\{Y = e\}$, a function of Y, so that

$$H(Y) = H(Y, E) = H(E) + H(Y|E) = h(\alpha) + (1 - \alpha)h(\pi), \tag{4}$$

because given $E = 0$ the output equals the input and given $E = 1$ it is constant.
Therefore $I(X;Y) = (1 - \alpha)h(\pi) \leq 1 - \alpha$, with equality at $\pi = \frac{1}{2}$:

$$C_{\mathrm{BEC}}(\alpha) = 1 - \alpha \quad \text{bits.} \tag{5}$$

The answer is rather natural: a fraction α of the bits are lost so at most a fraction $1-\alpha$ can be delivered, and the nonobvious part is that all of the surviving fraction can be used. With feedback this is easy: retransmit each erased bit until it gets through, which takes $1/(1-\alpha)$ uses per bit on average. Without feedback it is Shannon's theorem, and we will see that feedback cannot do better than $1-\alpha$ either.



binary symmetric channel | binary erasure channel | Z-channel

Figure 1: Three binary-input channels. Left, the binary symmetric channel with crossover probability p: errors are invisible to the receiver. Centre, the binary erasure channel: a fraction α of the bits are lost, visibly. Right, the Z-channel of Example 1.12: only the symbol 1 can be corrupted. The noisy typewriter of Example 1.5 is the 26-letter cycle in which each input goes to itself or to its successor with probability $\frac{1}{2}$.

Remark 1.8. *Z*-channel: sending a packet versus remaining silent. A lost packet looks like silence, but silence cannot create a packet. The interesting point is that silence is itself a valid symbol. If instead both bits were carried by packets and a missing packet produced a separate "erasure" output, we would obtain an erasure-channel model.

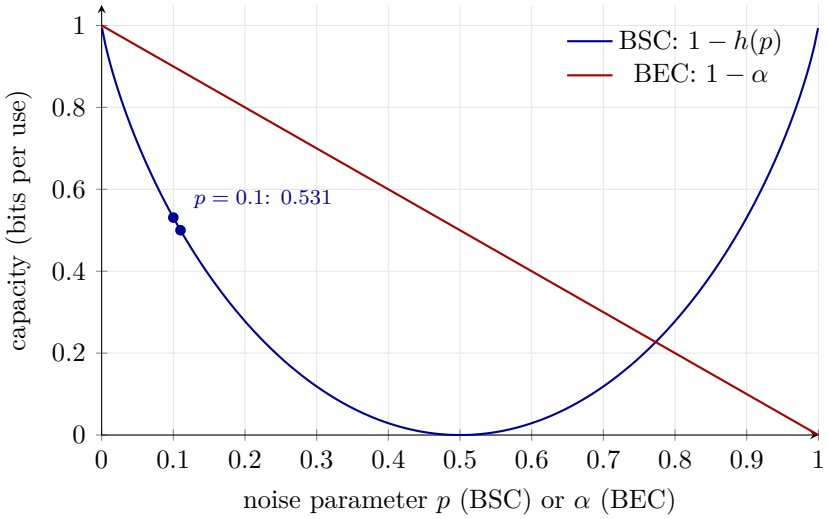


Figure 2: Capacity of the binary symmetric channel, $1-h(p)$, and of the binary erasure channel, $1-\alpha$, as functions of the noise parameter. Erasures cost less than errors: losing a fraction p of the bits visibly costs p bits per use, while flipping a fraction p invisibly costs $h(p) > p$, because the receiver must also learn where the flips were.

1.3 Symmetric channels

These are channels with a simple structure. In the BSC, the noisy typewriter and the noiseless channel the maximising input was uniform and the capacity was $\log|\mathcal{Y}|$ minus the entropy of a row of the transition matrix. That is a general phenomenon.

Definition 1.9 *(Symmetric and weakly symmetric channels).* A DMC is *symmetric* if every row of its transition matrix is a permutation of every other row and every column is a permutation of every other column.

It is *weakly symmetric* if every row is a permutation of every other row and all the column sums $\sum_x p(y|x)$ are equal.

Theorem 1.10 *(Capacity of weakly symmetric channels).* *For a weakly symmetric channel,*

$$C = \log|\mathcal{Y}| - H(\textit{row of the transition matrix}), \tag{6}$$

where the entropy is that of any row, all rows having the same entropy, and the capacity is attained by the uniform input distribution.

Proof. Let r be a row. Since every row is a permutation of r, so $H(Y|X=x) = H(r)$ for all x, and

$$I(X;Y) = H(Y) - H(Y|X) = H(Y) - H(r) \le \log|\mathcal{Y}| - H(r),$$

with equality if and only if Y is uniform. The uniform input makes Y uniform: with c the common column sum,

$$p(y) = \sum_x \frac{1}{|\mathcal{X}|}\, p(y|x) = \frac{c}{|\mathcal{X}|} \qquad \text{for every } y,$$

and this is the same for all y, hence equal to $1/|\mathcal{Y}|$. So the bound is attained. □

Example 1.11 *(A symmetric and a weakly symmetric channel).* The channel with transition matrix

$$p(y|x) = \begin{pmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{pmatrix}$$

is symmetric: the rows (indexed by x) are cyclic shifts of $(0.3, 0.2, 0.5)$ and so are the columns (indexed by y). Its capacity is $\log 3 - H(0.5, 0.3, 0.2) = 1.5850 - 1.4855 = 0.0995$ bits, a poor channel, since each row is not far from uniform.
The channel with transition matrix

$$p(y|x) = \begin{pmatrix} \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}$$

is not symmetric, since the first column is $(\frac{1}{3}, \frac{1}{3})$ and the second $(\frac{1}{6}, \frac{1}{2})$, but every column sums to $\frac{2}{3}$ and the rows are permutations of each other, so it is weakly symmetric, with $C = \log 3 - H(\frac{1}{3}, \frac{1}{6}, \frac{1}{2}) = 1.5850 - 1.4591 = 0.1258$ bits at the uniform input.

The channel $Y = X + Z \pmod c$ with Z independent of X on $\{0, \dots, c-1\}$ is symmetric, and (6) gives $C = \log c - H(Z)$: additive noise costs exactly its own entropy. The BSC is the case $c = 2$.

1.4 Z-channel, an asymmetric channel with a closed form

When the rows of the transition matrix are not permutations of one another the uniform input is in general not optimal and the maximisation in (1) has to be carried out. One channel small enough to do by hand shows what happens.

Example 1.12 *(The Z-channel).* $\mathcal{X} = \mathcal{Y} = \{0, 1\}$, the symbol 0 is always received correctly and the symbol 1 is received as 0 with probability p and as 1 with probability $1-p$ (Figure 1, right). Only one input is noisy, so the sender should use it less.
With $\pi = \mathbb{P}(X = 1)$ the output has $q = \mathbb{P}(Y = 1) = \pi(1-p)$, and

$$I(X;Y) = H(Y) - H(Y|X) = h(\pi(1-p)) - \pi\, h(p), \tag{7}$$

since $H(Y|X=0) = 0$ and $H(Y|X=1) = h(p)$.
The function is concave in π, and differentiating with $h'(q) = \log \frac{1-q}{q}$ gives the maximum value condition $(1-p)\log\frac{1-q}{q} = h(p)$, that is

$$q^* = \frac{1}{1+a}, \text{ with } a = 2^{h(p)/(1-p)}, \text{ and therefore } \pi^* = \frac{q^*}{1-p}. \tag{8}$$

Substituting into (7) and using $\log a = h(p)/(1-p)$, so that $\pi^* h(p) = q^* \log a$,

$$\begin{aligned} C &= h(q^*) - q^* \log a = \log(1+a) - \frac{a}{1+a}\log a - \frac{1}{1+a}\log a \\ &= \log\frac{1+a}{a} = \log\bigl(1 + (1-p)\,p^{p/(1-p)}\bigr), \end{aligned} \tag{9}$$

the last step because $-h(p)/(1-p) = \frac{p}{1-p}\log p + \log(1-p)$.
For $p = \frac{1}{2}$: $a = 4$, $q^* = 0.2$, $\pi^* = 0.4$ and $C = \log 1.25 = 0.3219$ bits, against $I = h(0.25) - 0.5 = 0.3113$ at the uniform input. The optimal sender uses the reliable symbol 60% of the time, and the gain over the uniform input is small, about 0.01 bit, which is typical: I is flat near its maximum (Figure 3). For $p = 0.1$, $C = 0.7628$ at $\pi^* = 0.4563$.

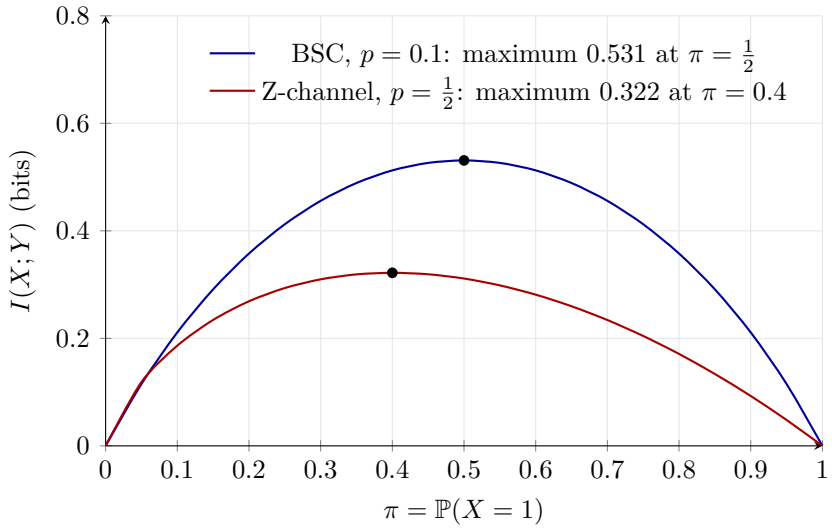


Figure 3: Mutual information as a function of the input distribution $\pi = \mathbb{P}(X = 1)$ for the BSC with $p = 0.1$ and for the Z-channel with $p = \frac{1}{2}$. Both curves are concave, as Proposition 1.13 guarantees; the symmetric channel peaks at $\pi = \frac{1}{2}$ and the asymmetric one at $\pi^* = 0.4$, where the reliable symbol is favoured.

1.5 Properties of the capacity

Proposition 1.13 *(Elementary properties of C).* *For every discrete memoryless channel:*
(i) $C \geq 0$.

(ii) $C \le \log|\mathcal{X}|$ *and* $C \le \log|\mathcal{Y}|$.
(iii) $I(X;Y)$ *is a continuous function of the input distribution* $p(x)$*, for fixed* $p(y|x)$.
(iv) $I(X;Y)$ *is a concave function of* $p(x)$*, for fixed* $p(y|x)$*. Consequently every local maximum of* I *over the input distributions is a global maximum, and the maximum in* (1) *is attained.*

Proof. (i) is $I(X;Y) \ge 0$.

(ii): $I(X;Y) \le H(X) \le \log|\mathcal{X}|$ and $I(X;Y) \le H(Y) \le \log|\mathcal{Y}|$.

(iii): $I(X;Y) = \sum_{x,y} p(x)p(y|x) \log \frac{p(y|x)}{\sum_{x'} p(x')p(y|x')}$ is a finite sum of continuous functions of the vector $p(x)$, the convention $0 \log 0 = 0$ making each term continuous at the boundary.

(iv): write $I(X;Y) = H(Y) - \sum_x p(x) H(Y|X = x)$. The second term is linear in $p(x)$. The output distribution $p(y) = \sum_x p(x)p(y|x)$ is linear in $p(x)$ and H is concave, so $H(Y)$ is a concave function of $p(x)$. A concave function on a convex compact set attains its maximum, and a local maximum of a concave function is global. □

2 Codes, rates and jointly typical sequences

The point of the session is to see, by counting, why 2^{nC} is the natural size of a good codebook, and to make the counting rigorous.

2.1 The communication system and a first count

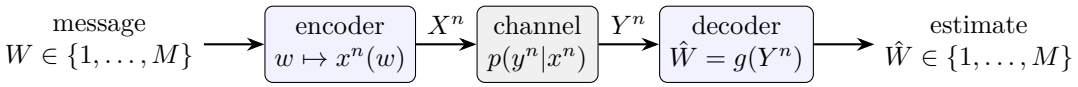


Figure 4: The communication system of these notes. A message W from $\{1, \dots, M\}$ is mapped by the encoder to a codeword $x^n(W)$, the channel produces Y^n with probability $p(y^n|x^n)$, and the decoder outputs a guess $\hat{W} = g(Y^n)$. The system fails when $\hat{W} \neq W$.

Figure 4 is the whole setting. The sender has M possible messages, uses the channel n times to convey one of them, and the receiver must decide which. Two inputs x^n and $\tilde{x}^n$ are *confusable* if some output y^n has positive probability under both; the sender's task is to choose M inputs that are as little confusable as possible, and the receiver's task is to make the best guess anyway. How large can M be?

Here is Shannon's heuristic count, made precise later in the session. Fix an input distribution $p(x)$ and imagine sending a typical input sequence x^n.

By the asymptotic equipartition property (AEP, Chapter 3 of Cover and Thomas), the output Y^n lands, with probability near one, in a set of about $2^{nH(Y|X)}$ sequences, each about equally likely: the *fan* of x^n.

The outputs of all inputs together occupy a set of about $2^{nH(Y)}$ typical sequences.

If we want the fans of our chosen inputs to be disjoint, so that every output points *to one single input*, we can fit at most

$$\frac{2^{nH(Y)}}{2^{nH(Y|X)}} = 2^{n(H(Y)-H(Y|X))} = 2^{nI(X;Y)} \tag{10}$$

of them (Figure 5).

Choosing $p(x)$ to maximise $I(X;Y)$ gives 2^{nC}:

capacity is the exponent of the number of **distinguishable inputs**.

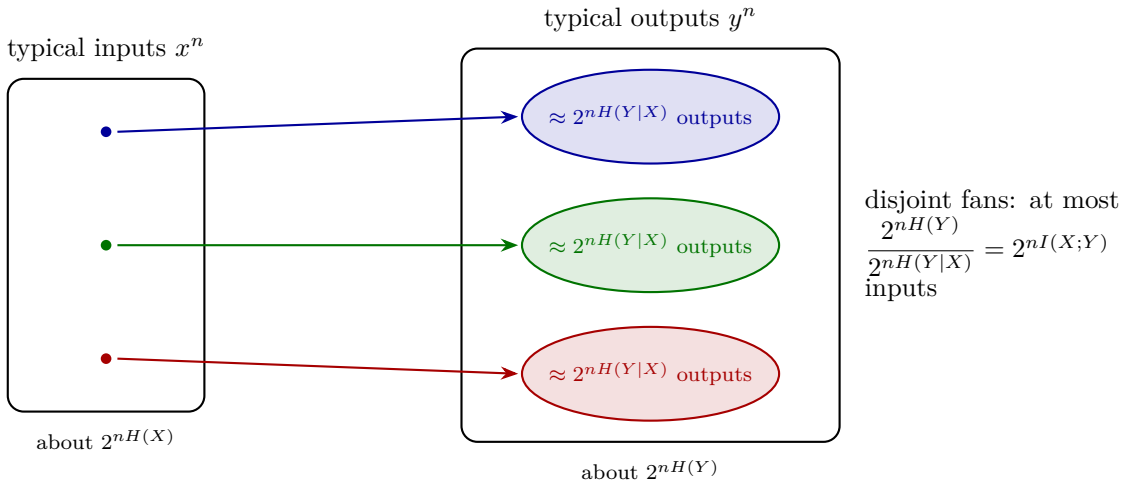


Figure 5: Shannon's count. Each typical input sequence fans out to about $2^{nH(Y|X)}$ likely outputs inside the set of about $2^{nH(Y)}$ typical outputs. At most $2^{n(H(Y)-H(Y|X))} = 2^{nI(X;Y)}$ fans fit without overlapping, so at most 2^{nI} inputs can be told apart from the output.

Example 2.1 *(The count for the BSC and for the noisy typewriter).* For the BSC with $p = 0.1$, $n = 100$ and uniform input, $H(Y|X) = h(0.1) = 0.4690$, so each input fans out to about $2^{46.9}$ outputs, out of 2^{100} typical outputs: at most $2^{53.1}$ fans fit, which is 2^{nC} with $C = 0.5310$. The count suggests that about 53 bits, and no more, can be carried by 100 uses of a channel that flips one bit in ten. For the noisy typewriter the count is exact: every input sequence fans out to exactly 2^n outputs, all typical outputs are the 26^n letter strings, and the 13^n sequences over the alternate letters have disjoint fans that cover everything, so $2^{n \log 26}/2^n = 13^n = 2^{n \log 13}$ inputs are distinguishable with no error at all.

2.2 Codes, error probabilities and rates

A sequence of n uses of a DMC defines a new DMC.

Definition 2.2 *(The nth extension of a DMC).* The *nth extension* of the DMC $(\mathcal{X}, p(y|x), \mathcal{Y})$ is the channel $(\mathcal{X}^n, p(y^n|x^n), \mathcal{Y}^n)$ with

$$p(y_k \mid x^k, y^{k-1}) = p(y_k \mid x_k), \qquad k = 1, \ldots, n. \tag{11}$$

If the channel is used *without feedback*, that is, if the inputs do not depend on past outputs, $p(x_k \mid x^{k-1}, y^{k-1}) = p(x_k \mid x^{k-1})$, then

$$p(y^n \mid x^n) = \prod_{i=1}^{n} p(y_i \mid x_i). \tag{12}$$

Equation (12) is used throughout Sessions 2 to 5 and is exactly what fails with feedback, the subject of the last part of Session 4. Its proof is the chain rule: $p(y^n|x^n) = \prod_k p(y_k|x^n, y^{k-1})$, and without feedback the future inputs $x_{k+1}, \ldots, x_n$ are conditionally independent of y_k given x^k, y^{k-1}, so the factor reduces to (11).

Suppose we have M possible messages to send through the channel.

Definition 2.3 *(Code, error probabilities, rate).* An (M, n) *code* for the channel $(\mathcal{X}, p(y|x), \mathcal{Y})$ consists of

- an index set $\{1, \dots, M\}$,
- an *encoding function* $x^n : \{1, \dots, M\} \to \mathcal{X}^n$ whose values $x^n(1), \dots, x^n(M)$ are the *codewords* and together form the *codebook*,
- a deterministic *decoding function* $g : \mathcal{Y}^n \to \{1, \dots, M\}$.

Its *conditional probability of error* given that message w was sent is

$$\lambda_w = \mathbb{P}(g(Y^n) \neq w \mid X^n = x^n(w)) = \sum_{y^n} p(y^n \mid x^n(w))\, \mathbf{1}\{g(y^n) \neq w\}, \tag{13}$$

its *maximal probability of error* is $\lambda^{(n)} = \max_w \lambda_w$,

and its *average probability of error* is $P_e^{(n)} = \frac{1}{M} \sum_{w=1}^{M} \lambda_w$.

Its *rate* is

$$R = \frac{\log M}{n} \quad \text{bits per channel use.} \tag{14}$$

If the message W is drawn uniformly from $\{1, \dots, M\}$ then $P_e^{(n)} = \mathbb{P}(\hat{W} \neq W)$ is literally the probability of error of the system; for any other message distribution $P_e^{(n)}$ is just the arithmetic mean of the λ_w, and $\lambda^{(n)}$ is a guarantee that holds whatever the sender's distribution.

Obviously $P_e^{(n)} \leq \lambda^{(n)}$. The maximal criterion is the honest one, since a code with small $P_e^{(n)}$ could still fail badly on a few messages; Session 3 shows that the two criteria give the same capacity, because **the bad half of a codebook can be discarded at negligible cost in rate.**

Example 2.4 *(Two codes for the BSC).* On the BSC with $p = 0.1$ the *uncoded* scheme sends n message bits as they are: an $(2^n, n)$ code of rate 1 with g the identity. Every message has $\lambda_w = 1 - (1-p)^n$, which is 0.4095 for $n = 5$ and tends to 1: full rate, no reliability.
The *repetition code* of length 3 sends `000` for message 1 and `111` for message 2, an $(2, 3)$ code of rate $\frac{1}{3}$, decoded by majority vote. An error occurs when two or three bits flip, so $\lambda_1 = \lambda_2 = 3p^2(1-p) + p^3 = 0.028$, and $\lambda^{(n)} = P_e^{(n)} = 0.028$. Repeating 5, 7 or 9 times gives 0.0086, 0.0027 and 0.0009 at rates $\frac{1}{5}, \frac{1}{7}, \frac{1}{9}$.
Reliability improves only as the rate goes to zero. Shannon's theorem says that this trade-off is an artefact of a poor code: at any rate below $C = 0.531$, arbitrarily small error is possible.

Definition 2.5 *(Achievable rate, operational capacity).* A rate $R \geq 0$ is *achievable* for a channel if there exists a sequence of $(\lceil 2^{nR} \rceil, n)$ codes, $n = 1, 2, \dots$, whose maximal probability of error $\lambda^{(n)}$ tends to 0 as $n \to \infty$. The *operational capacity* of the channel is the supremum of the achievable rates.

Remark 2.6. By Convention 0.1 we write $(2^{nR}, n)$ for $(\lceil 2^{nR} \rceil, n)$. Achievability is a statement about a sequence of codes of growing block length: the theory does not promise any single finite code with error exactly zero, only that the error can be made as small as desired by coding over long enough blocks. Example 2.4 shows that $R = 0$ is achievable in a rather silly way; the question is how large R can be.

2.3 Jointly typical sequences

The decoder of Session 3 will look for a codeword that "fits", that is is "jointly typical" to the received sequence. The notion of joint typicality extends the typical set of the AEP from one sequence to a pair.

Throughout, (X^n, Y^n) is drawn i.i.d. from a joint distribution $p(x, y)$ on $\mathcal{X} \times \mathcal{Y}$, so $p(x^n, y^n) = \prod_i p(x_i, y_i)$, and $p(x^n)$, $p(y^n)$ are the marginals.

Definition 2.7 *(Jointly typical set).* For $\varepsilon > 0$, the set $A_\varepsilon^{(n)}$ of *jointly ε-typical* pairs with respect to $p(x, y)$ consists of the pairs $(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n$ such that

$$\begin{aligned} \left| -\tfrac{1}{n} \log p(x^n) - H(X) \right| &< \varepsilon, \\ \left| -\tfrac{1}{n} \log p(y^n) - H(Y) \right| &< \varepsilon, \\ \left| -\tfrac{1}{n} \log p(x^n, y^n) - H(X,Y) \right| &< \varepsilon. \end{aligned} \tag{15}$$

In words: x^n is typical for X, y^n is typical for Y, and the pair is typical for the joint law, all three empirical entropies being within ε of the true ones.

The third condition is the one that carries the dependence: a typical x^n and a typical y^n need not form a *jointly typical* pair.

As before, we have now an asymptotic equipartition property for *jointly typical* pairs:

Theorem 2.8 *(Joint AEP).* *Let (X^n, Y^n) be i.i.d. with law $p(x, y)$, and let $\varepsilon > 0$.*

(i) $\mathbb{P}\big((X^n, Y^n) \in A_\varepsilon^{(n)}\big) \to 1$ *as* $n \to \infty$.

(ii) $|A_\varepsilon^{(n)}| \le 2^{n(H(X,Y)+\varepsilon)}$.

(iii) If $(\tilde{X}^n, \tilde{Y}^n) \sim p(x^n)p(y^n)$, that is, $\tilde{X}^n$ and $\tilde{Y}^n$ are independent with the same marginals as X^n and Y^n, then

$$\mathbb{P}\big((\tilde{X}^n, \tilde{Y}^n) \in A_\varepsilon^{(n)}\big) \le 2^{-n(I(X;Y)-3\varepsilon)}, \tag{16}$$

and for n large enough also $\mathbb{P}\big((\tilde{X}^n, \tilde{Y}^n) \in A_\varepsilon^{(n)}\big) \ge (1-\varepsilon)\, 2^{-n(I(X;Y)+3\varepsilon)}$.

Proof. (i): $-\frac{1}{n} \log p(X^n) = \frac{1}{n} \sum_i \log \frac{1}{p(X_i)}$ is an average of i.i.d. random variables with mean $H(X)$, so by the weak law of large numbers it converges to $H(X)$ in probability.

The same holds for $-\frac{1}{n} \log p(Y^n) \to H(Y)$ and $-\frac{1}{n} \log p(X^n, Y^n) \to H(X, Y)$.

Hence there is n_0 such that for $n \ge n_0$ each of the three events in (15) fails with probability smaller than $\varepsilon/3$, and by the union bound all three hold with probability greater than $1 - \varepsilon$.

(ii): every pair in $A_\varepsilon^{(n)}$ has $p(x^n, y^n) > 2^{-n(H(X,Y)+\varepsilon)}$ by the third condition, so

$$1 \ge \sum_{(x^n, y^n) \in A_\varepsilon^{(n)}} p(x^n, y^n) \ge |A_\varepsilon^{(n)}|\, 2^{-n(H(X,Y)+\varepsilon)}.$$

(iii): for an independent pair in $A_\varepsilon^{(n)}$ the first two conditions give $p(x^n) \le 2^{-n(H(X)-\varepsilon)}$ and $p(y^n) \le 2^{-n(H(Y)-\varepsilon)}$, so, using (ii),

$$\begin{aligned} \mathbb{P}\big((\tilde{X}^n, \tilde{Y}^n) \in A_\varepsilon^{(n)}\big) &= \sum_{(x^n, y^n) \in A_\varepsilon^{(n)}} p(x^n) p(y^n) \\ &\le 2^{n(H(X,Y)+\varepsilon)}\, 2^{-n(H(X)-\varepsilon)}\, 2^{-n(H(Y)-\varepsilon)} = 2^{-n(I(X;Y)-3\varepsilon)}, \end{aligned}$$

since $H(X) + H(Y) - H(X, Y) = I(X; Y)$.

For the lower bound, (i) gives, for n large, $1 - \varepsilon \le \sum_{A_\varepsilon^{(n)}} p(x^n, y^n) \le |A_\varepsilon^{(n)}| 2^{-n(H(X,Y)-\varepsilon)}$, and hence $|A_\varepsilon^{(n)}| \ge (1-\varepsilon) 2^{n(H(X,Y)-\varepsilon)}$, and then the same computation with the reverse inequalities give $p(x^n) \ge 2^{-n(H(X)+\varepsilon)}$, $p(y^n) \ge 2^{-n(H(Y)+\varepsilon)}$ and the claim follows. □

The theorem is the intuitive count of Section 2.1 made exact.

There are about $2^{nH(X)}$ typical inputs and $2^{nH(Y)}$ typical outputs, so about $2^{n(H(X)+H(Y))}$ pairs of typical sequences, but only about $2^{nH(X,Y)}$ of them are jointly typical: a fraction $2^{-nI(X;Y)}$.

Part (iii) says that a randomly chosen unrelated pair is jointly typical with probability about 2^{-nI}.

So if a decoder declares "codeword i was sent" whenever $(x^n(i), y^n)$ is jointly typical, a codeword is incorrectly identified with probability about 2^{-nI}, and about 2^{nI} codewords can be accommodated before false identifications become likely.

Example 2.9 *(Joint typicality for the BSC, exactly).* For the BSC with crossover p and uniform input, everything can be computed exactly, because the three conditions of (15) depend on the pair (x^n, y^n) only through the number k of positions where $x_i \neq y_i$.
Indeed $p(x^n) = p(y^n) = 2^{-n}$, so the first two conditions hold for every pair, and $p(x^n, y^n) = 2^{-n} p^k (1-p)^{n-k}$, so the third reads

$$\left| \tfrac{k}{n} \log \tfrac{1}{p} + (1 - \tfrac{k}{n}) \log \tfrac{1}{1-p} - h(p) \right| < \varepsilon : \tag{17}$$

the fraction of disagreements must be close to p.
Take $p = 0.1$ and $\varepsilon = 0.1$. Check that for $n = 100$ the window is $7 \leq k \leq 13$; the probability that the true pair (X^n, Y^n) falls in it is $\sum_{k=7}^{13} \binom{100}{k} 0.1^k 0.9^{100-k} = 0.7590$; the jointly typical set has $2^{100} \sum_{k=7}^{13} \binom{100}{k} = 2^{152.89}$ elements, below the bound $2^{n(H(X,Y)+\varepsilon)} = 2^{156.90}$; and an independent pair $(\tilde{X}^n, \tilde{Y}^n)$, which disagrees in a Binomial$(100, \frac{1}{2})$ number of places, is jointly typical with probability $\sum_{k=7}^{13} \binom{100}{k} 2^{-100} = 6.6 \times 10^{-15} = 2^{-47.11}$, far below the guaranteed $2^{-n(I-3\varepsilon)} = 2^{-23.10}$ and close to the heuristic $2^{-nI} = 2^{-53.1}$. For $n = 1000$ the window is $69 \leq k \leq 131$, the true pair is jointly typical with probability 0.9991, and an independent pair with probability $2^{-444.3}$.

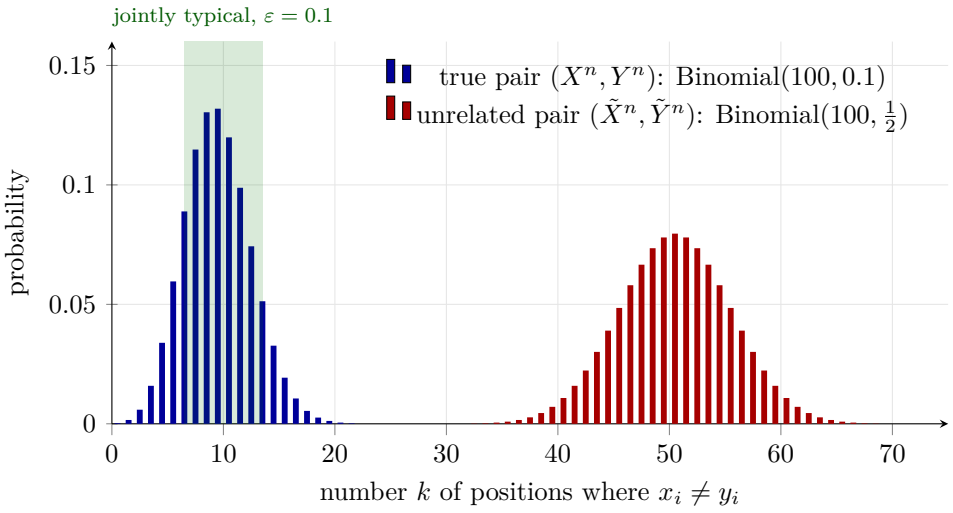


Figure 6: Number of disagreements between x^n and y^n for $n = 100$ on the BSC with $p = 0.1$: for the pair actually produced by the channel (left, Binomial$(100, 0.1)$) and for an unrelated pair (right, Binomial$(100, \frac{1}{2})$). The shaded band $7 \leq k \leq 13$ is the jointly typical window of Example 2.9 with $\varepsilon = 0.1$; the unrelated pair lands in it with probability 2^{-47}.

3 The channel coding theorem: achievability

The theorem of this session is the reason information theory exists as a subject. It says that the number computed in Session 1 is achievable: for every rate below C there are codes whose probability of error, on every message, goes to zero as the block length grows.

The claim sounds paradoxical, because a channel that corrupts symbols cannot be made to stop corrupting them, and any correction procedure is itself carried out on corrupted symbols.

Shannon's channel coding theorem has three ingredients: allow the error to be *small but not zero*, use the channel many times so that the *law of large numbers* acts, and prove that a good code exists by showing that the *average* code is good.

3.1 The theorem

Theorem 3.1 *(Channel coding theorem).* *For a discrete memoryless channel with capacity* $C = \max_{p(x)} I(X;Y)$,

- *every rate* $R < C$ *is achievable: there exists a sequence of* $(2^{nR}, n)$ *codes with maximal probability of error* $\lambda^{(n)} \to 0$.
- *conversely, every sequence of* $(2^{nR}, n)$ *codes with* $\lambda^{(n)} \to 0$ *has* $R \le C$.

The first statement is proved in this session and the second, the converse, in Session 4.

Together they say that the **operational capacity** of Definition 2.5 equals the **information capacity** of Definition 1.2, which is why the two are simply called the capacity.

3.2 The construction

Fix a rate $R < C$ and an input distribution $p(x)$, to be chosen later.

The code is built as follows.

(1) *Random codebook.* Draw 2^{nR} codewords independently, each with i.i.d. symbols from $p(x)$: the codebook is the $2^{nR} \times n$ array

$$\mathcal{C} = \begin{pmatrix} x_1(1) & x_2(1) & \cdots & x_n(1) \\ \vdots & \vdots & & \vdots \\ x_1(2^{nR}) & x_2(2^{nR}) & \cdots & x_n(2^{nR}) \end{pmatrix}, \qquad \mathbb{P}(\mathcal{C}) = \prod_{w=1}^{2^{nR}} \prod_{i=1}^{n} p(x_i(w)). \tag{18}$$

(2) The codebook $\mathcal{C}$ and the channel law $p(y|x)$ are known to sender and receiver.

(3) A message W is drawn uniformly from $\{1, \dots, 2^{nR}\}$ and the codeword $X^n(W)$, the Wth row of $\mathcal{C}$, is sent.

(4) The receiver observes Y^n with probability $p(y^n|x^n(W)) = \prod_i p(y_i|x_i(W))$.

(5) *Joint-typicality decoding.* The receiver declares $\hat{W} = w$ if $(x^n(w), Y^n) \in A_\varepsilon^{(n)}$ and no other index $w' \neq w$ has $(x^n(w'), Y^n) \in A_\varepsilon^{(n)}$.
If no index or more than one index qualifies, the receiver declares an error (say, outputs 0).

The decoder that minimises the probability of error is the maximum-likelihood decoder, which picks the w maximising $p(y^n|x^n(w))$, but it is hard to analyse. Joint-typicality decoding is weaker, since it may refuse to decide where maximum likelihood would decide correctly, but it is easy to analyse and turns out to be good enough for every rate below capacity.

3.3 The error analysis

Let $\mathcal{E} = \{\hat{W} \neq W\}$ be the error event. We compute the probability of $\mathcal{E}$ averaged over the random message *and* over the random codebook:

$$\mathbb{P}(\mathcal{E}) = \sum_{\mathcal{C}} \mathbb{P}(\mathcal{C})\, P_e^{(n)}(\mathcal{C}) = \sum_{\mathcal{C}} \mathbb{P}(\mathcal{C})\, \frac{1}{2^{nR}} \sum_{w=1}^{2^{nR}} \lambda_w(\mathcal{C}) = \frac{1}{2^{nR}} \sum_{w=1}^{2^{nR}} \sum_{\mathcal{C}} \mathbb{P}(\mathcal{C}) \lambda_w(\mathcal{C}). \tag{19}$$

The inner sum does not depend on w: the codebook distribution (18) is invariant under permuting rows, and the decoding rule treats all indices alike, so the average conditional error of message w equals that of message 1.

Hence

$$\mathbb{P}(\mathcal{E}) = \sum_{\mathcal{C}} \mathbb{P}(\mathcal{C}) \lambda_1(\mathcal{C}) = \mathbb{P}(\mathcal{E} \mid W = 1), \tag{20}$$

and we may assume that message 1 was sent. Define the events

$$E_w = \{(X^n(w), Y^n) \in A_\varepsilon^{(n)}\}, \qquad w = 1, \dots, 2^{nR}, \tag{21}$$

where Y^n is the output produced by $X^n(1)$.

Message 1 is decoded wrongly only if E_1 fails (the true codeword does not look typical with the output) or some E_w with $w \geq 2$ occurs (an impostor looks typical).

By the union bound,

$$\mathbb{P}(\mathcal{E} \mid W = 1) = \mathbb{P}\Big(E_1^c \cup \bigcup_{w=2}^{2^{nR}} E_w \;\Big|\; W = 1\Big) \leq \mathbb{P}(E_1^c \mid W = 1) + \sum_{w=2}^{2^{nR}} \mathbb{P}(E_w \mid W = 1). \tag{22}$$

The two kinds of term are controlled by the two halves of the joint AEP.

The true codeword. $(X^n(1), Y^n)$ is i.i.d. with law $p(x)p(y|x)$, so by Theorem 2.8(i), $\mathbb{P}(E_1^c \mid W = 1) \leq \varepsilon$ for n large enough.

The impostors. For $w \neq 1$ the codeword $X^n(w)$ was drawn independently of $X^n(1)$, hence independently of Y^n, and it has the marginal law $p(x^n)$ while Y^n has the marginal law $p(y^n)$.

So $(X^n(w), Y^n)$ is exactly the independent pair of Theorem 2.8(iii), and $\mathbb{P}(E_w \mid W = 1) \leq 2^{-n(I(X;Y)-3\varepsilon)}$.

Putting the two into (22),

$$\mathbb{P}(\mathcal{E}) \leq \varepsilon + (2^{nR} - 1)\, 2^{-n(I(X;Y)-3\varepsilon)} \leq \varepsilon + 2^{-n(I(X;Y)-R-3\varepsilon)} \leq 2\varepsilon \tag{23}$$

for n large enough, provided $R < I(X;Y) - 3\varepsilon$.

Since $\varepsilon > 0$ was arbitrary, we have shown: *if $R < I(X;Y)$, then for every $\varepsilon > 0$ and all large n the probability of error, averaged over messages and over codebooks, is at most 2ε.*

3.4 From the average to a code

Three selections turn the average statement into the theorem.

Choose the input distribution. Take $p(x) = p^*(x)$, a distribution attaining C, which exists by Proposition 1.13. Then the condition $R < I(X;Y)$ becomes $R < C$.

Choose the codebook. An average over codebooks that is at most 2ε forces at least one codebook $\mathcal{C}^*$ with $P_e^{(n)}(\mathcal{C}^*) \leq 2\varepsilon$: a random variable cannot exceed its mean everywhere.

Such a codebook can in principle be found by exhaustive search over all $(2^{nR}, n)$ codes; the proof does not tell us which one it is, **only that it exists**.

Discard the worst half. $P_e^{(n)}(\mathcal{C}^*) = 2^{-nR} \sum_w \lambda_w(\mathcal{C}^*) \leq 2\varepsilon$ is an average of 2^{nR} nonnegative numbers, so at most half of them can exceed 4ε, otherwise the average would exceed 2ε.

Keep the 2^{nR-1} codewords with the smallest λ_w and reindex them: the resulting code has $2^{nR-1} = 2^{n(R-1/n)}$ codewords and maximal probability of error $\lambda^{(n)} \leq 4\varepsilon$.

The rate has dropped from R to $R - \frac{1}{n}$, which is negligible for large n, and any rate below C is still covered by starting slightly higher.

This proves the achievability half of Theorem 3.1.

3.5 What the proof gives

Random coding is a proof technique, not a transmission scheme. Randomness enters only to make the average computable. The code that is finally used is a fixed deterministic codebook $\mathcal{C}^*$ agreed on in advance. Step (2) of the construction costs nothing: sender and receiver do not transmit the codebook over the channel, they both know "the best $(2^{nR}, n)$ code for this channel", a description of constant length.

Existence without construction. The proof shows that almost every codebook is good, and gives no way of writing one down except by drawing it at random and checking. A random codebook of block length n has 2^{nR} codewords with no structure, so encoding needs a table with 2^{nR} rows and decoding by table lookup needs, for each received y^n, a comparison with every codeword: exponential in n. The theorem therefore poses rather than solves the engineering problem.

References

[1] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory.* Wiley-Interscience, Hoboken, NJ, 2nd edition, 2006. The course textbook.